raccoon: A package to correct for wiggles in the JWST NIRSpec integral field spectroscopy

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Summary

raccoon is a Python package designed to correct resampling noise — commonly referred to as "wiggles" — in the reduced spectra from the JWST Near Infrared Spectrograph's (NIRSpec) integral field spectroscopy (IFS) mode. These wiggles arise as artifacts during the resampling of the 2D raw data, affected by undersampling of the point spread function (PSF), into 3D datacubes. The standard JWST data reduction pipeline does not correct for this noise. The wiggle artifacts can significantly degrade the scientific usability of the data, particularly at the pixel level, undermining the exquisite spatial resolution of NIRSpec. raccoon provides an empirical correction by modeling and removing these artifacts, thereby restoring the fidelity of the extracted spectra. Whereas previously available tools for this purpose first extract a "wiggle signal" from the data and then model it through a fast Fourier transform, raccoon models the wiggles as a sinusoidal chirp function impacting a template spectrum that is directly fit to the original data, without an intermediate extraction step. As a result, raccoon robustly captures the global characteristics of the wiggles while avoiding potential impacts from local imperfections in an extracted wiggle signal.

Statement of need

The JWST NIRSpec's IFS mode (Böker et al., 2023) enables spatially resolved infrared spectroscopy of astronomical sources with an unprecedented combination of signal-to-noise ratio, redshift coverage, and spatial resolution. However, it suffers from heavy under-sampling of the point-spread function, leading to resampling noise that manifests as low-frequency sinusoidal artifacts known as "wiggles." These artifacts can significantly distort the overall spectral shape, bias line measurements, and compromise kinematic analyses at the single-pixel level, thereby limiting the scientific potential of the NIRSpec IFU data. raccoon provides a user-friendly, robust, and computationally efficient solution to identify and remove these artifacts, enabling precise studies of galaxy kinematics and early-Universe phenomena that would otherwise be hindered by the presence of wiggles. raccoon has already been used in scientific publications to facilitate robust measurements of stellar kinematics from the JWST NIRSpec spectra (Shajib et al., 2025; TDCOSMO Collaboration, 2025).

Functionality

Since the wiggles in the single-spaxel spectra in the reduced JWST NIRSpec datacube are artifacts due to resampling noise from PSF undersampling, dithering can be used in principle to mitigate the issue of PSF undersampling and thus the wiggles. However, the typically adopted

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4-point dither pattern is still insufficient to recover the optimal sampling and completely eliminate the wiggles in the rectified datacube (Law et al., 2023). The wiggles are washed out in spectra summed from multiple spaxels within a sufficiently large aperture (illustrated in Fig. 1). This provides a basis for making an empirical correction for the wiggles by comparing the single-spaxel spectrum to the one summed within an aperture around it. This principle was also used by previous corrective algorithms such as Perna et al. (2023) and in the Python routine `WICKED` (Dumont et al., 2025).

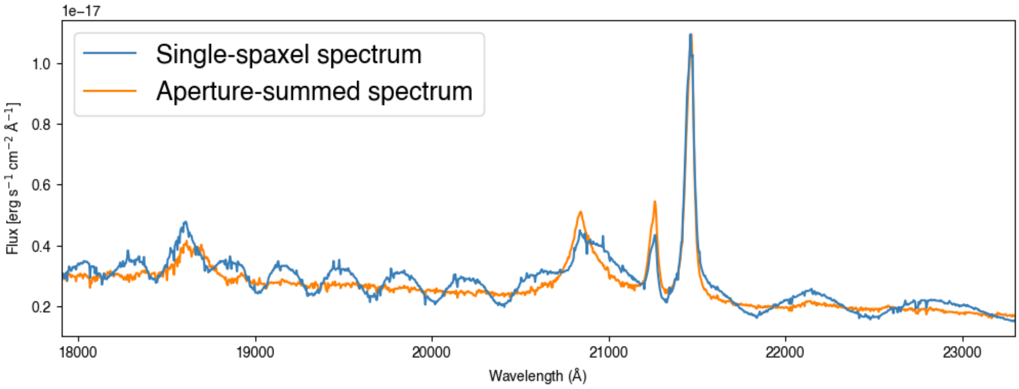

Figure 1: Illustration of the wiggles in the single-spaxel spectrum (blue), which is a manifestation of the resampling noise in the standard-pipeline-reduced datacube due to PSF undersampling. The orange spectrum shows the aperture-summed spectrum within a 4-spaxel radius. The illustrated data are of an active galactic nucleus (AGN) from Perna et al. (2023).

In `raccoon`, the wiggle is modeled as a sinusoidal chirp function

$$W(\lambda) = 1 + A(\lambda)\left[\sin(\phi_\lambda) + k_1 \sin^2(\phi_\lambda) + k_2 \sin(3\phi_\lambda)\right],$$

where $A(\lambda)$ is the wavelength-dependent amplitude and $\phi_\lambda = \lambda\, k(\lambda) + \phi_0$ is the wavelength-dependent phase term. The wiggle's peaks and troughs can be asymmetrically and symmetrically sharpened (or, de-sharpened) by freely varying the k_1 and k_2 parameters, respectively.

Given this model for the wiggle, a single-spaxel spectrum $D(\lambda)$ is modeled with

$$M(\lambda) = W(\lambda)\, T(\lambda),$$

where $T(\lambda)$ is a template for the correct spectrum devoid of the wiggles. In `raccoon`, this template is constructed based on the circular aperture-summed spectrum $C(\lambda)$. The template can also optionally include a spectrum $S(\lambda)$ summed from a shell or annulus centered on the corresponding spaxel, following Dumont et al. (2025). Including the shell-summed spectrum in the template can account for changes in the line shape between the single-spaxel spectrum and the aperture-summed one, for example, in the case of lines broadened by stellar kinematics that can vary across the galaxy (Dumont et al., 2025). The aperture radius and the inner and outer radii of the shell are to be adjusted by the user, as the appropriate values depend on the source morphology and the astronomical scene. The template is constructed as

$$T(\lambda) = c_1\, C(\lambda) + c_2\, S(\lambda) + c_3\, \lambda^b + \sum_{n=0}^{N} p_n\, \lambda^n,$$

where $c_1, c_2, c_3, p_0, \ldots, p_N$ are linear coefficients. Here, the power-law plus polynomial $c_3\, \lambda^b + \sum_{n=0}^{N} p_n\, \lambda^n$ on the right-hand side models the difference in the continuum between the single-spaxel spectrum and $c_1\, C(\lambda) + c_2\, S(\lambda)$.

The functions $A(\lambda)$ and ϕ_λ are modeled with B-splines, with the number of knots adjustable by the user. `raccoon` provides a functionality for the user to determine the appropriate number of knots using model selection criteria such as the Bayesian information criterion (BIC) or the minimum *a posteriori* chi-squared metric ($\chi^2_{\rm MAP}$). The best-fit values for the linear coefficients $c_1, c_2, c_3, p_0, \dots, p_N$ and non-linear parameters (b and the coefficients of the B-spline basis functions) are determined by minimizing the χ^2 function:

$$\chi^2 = \sum_i \frac{(D_i - M_i)^2}{\sigma_i^2},$$

where the index i runs across the wavelength pixels and σ_i is the associated noise level. Figure 2 illustrates an example of the fitted model to a given spectrum (of an AGN). In this example, raccoon's robust performance is demonstrated in fitting the given spectrum while modeling the wiggle signal (Figure 3). The user can mask out regions of the spectrum that have strong features potentially impacting the quality of the fit, or optionally adopt outlier rejection in the fit using the false discovery rate method (Benjamini & Hochberg, 2018) or sigma-clipping.

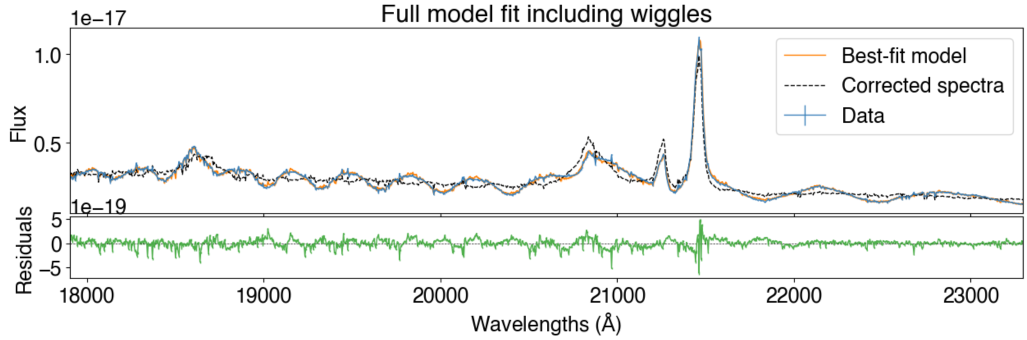

Figure 2: Modeling of the full spectrum (blue in the top panel) based on the template spectra and the wiggles impacting it. The illustrated spectrum is the same one from Fig **??**. The best-fit model is shown in orange and the wiggle-corrected spectrum is shown in black. The bottom panel illustrates the residuals (green) between the original data and the best-fit model.

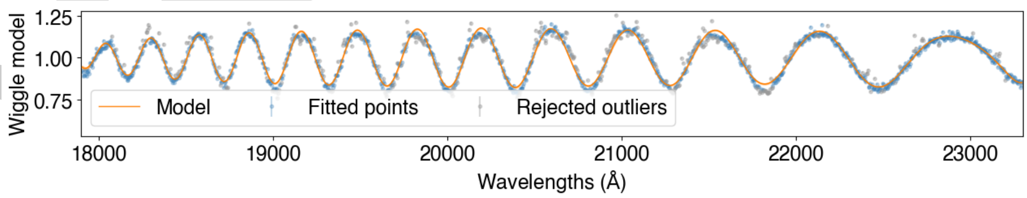

Figure 3: Illustration of the extracted and modeled wiggle signal. The illustrated data points represent $D(\lambda)/M(\lambda)$ based on the best-fit $M(\lambda)$, and the orange line illustrates the best-fit wiggle model $W(\lambda)$. The grey points are rejected outliers using the false discovery rate method, and the blue points mark the wavelengths where the data were fit to the model.

raccoon additionally allows the user to easily loop through multiple spaxels within the datacube, optionally within a user-specified region. The user can also set a detection threshold for the wiggle signal before making a correction on the given spectra.

There are several advantages of raccoon over previously available scripts and routines. Both previous tools (Dumont et al., 2025; Perna et al., 2023) first extract the wiggle signal by comparing the single-spaxel spectrum with the template spectrum and then model the low-frequency behavior of the wiggles through a fast Fourier transform. raccoon fits the model including the wiggles directly to the data, bypassing the need for an intermediate extraction step. Furthermore, raccoon employs a single parametric model that is continuous through the fitted wavelength range. This combination of differences makes raccoon less susceptible to local imperfections (potentially present in an extracted wiggle signal) or gaps in the wavelength range, either excluded through masking of strong spectral features or through outlier rejection. Another notable difference is that previously available tools make an additive correction, whereas `raccoon` makes a multiplicative one. However, since both cases obtain the necessary correction factor or term empirically from the data, this difference should not put either of them at a disadvantage at a practical level. However, the wiggles manifest as a multiplicative effect on the extracted spectra (Law et al., 2023), which is the reason `raccoon` models them as such. Furthermore, uniquely among its peers, `raccoon` is installable through the `pip` command and thus is more user-friendly in its portability and flexibility of use.

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